Prosthetic feet

SACH foot (Otto Bock)





Trailblazer® MA Foot



Pathfinder II Foot





Single Axis foot (WillowWood)



http://newsfeed.time.com/2013/02/23/sum-of-the-parts-prosthetics-through-the-centuries/slide/inside-foot/

Prosthetic feet



Fusion® Foot



Pathfinder II Foot



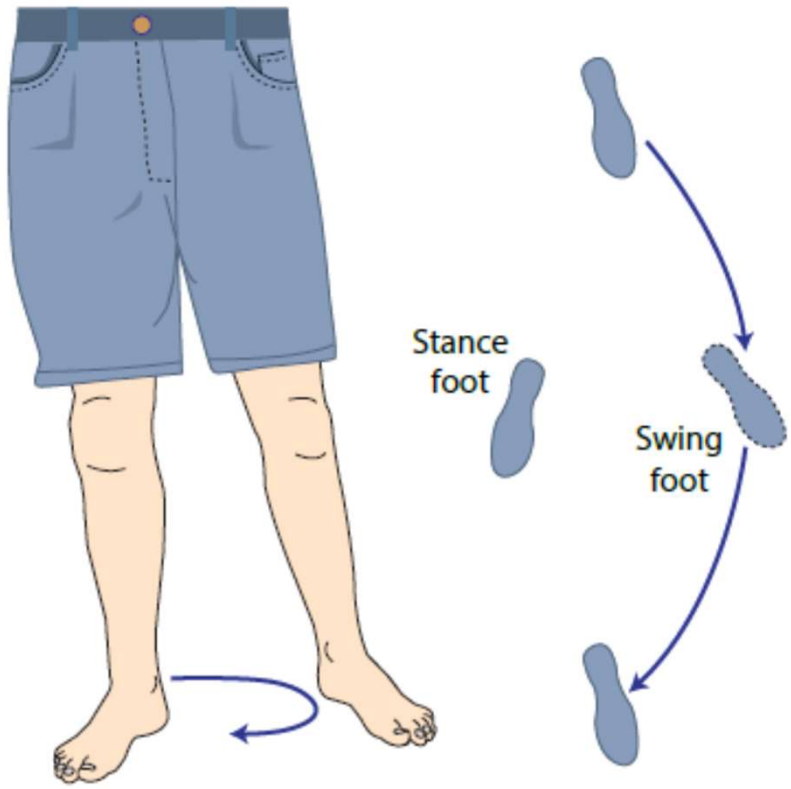
Trailblazer® MA Foot

Role of a lower limb prosthesis



- Provide stability in standing and walking
- Shock absorption
- Provide ability to sit comfortably
- Enable forward motion of leg to prepare for the next step
- Assist in push-off
- Control pendular swing
- Adjust to different walking speeds
- Reduce effort needed to walk
- Restore independence
- Cater to other specialized functional needs - swimming, sprinting

Walking with a locked knee



Abnormal gait

Increased energy expenditure
[Mattsson and Brostromm,1990]

Damage to other joints in the long run [CIR, 2006]

EZ-LOK



User-friendly alternative to drop lock

Easy operation for unilateral and bilateral KAFO users

Improved safety

Smooth contours

Increased knee flexion

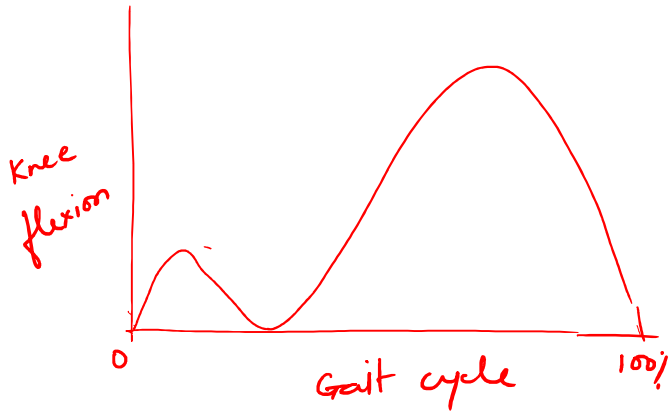
Design Patent no. 290752

Courtesy Dr. Ganesh Bapat, Alumnus R2D2, IITM

Prosthetic knee – design considerations

Compensate for missing muscles and normal control mechanisms

- Stability in stance phase
  - Braking mechanism
  - Geometric stability
  - Microprocessor-controlled
- Swing-phase control
  - Accommodate different walking speeds
  - Assist extension of the shank
  - Provide toe clearance
- Shock absorption
  - Controlled stance flexion during Loading Response
  - Contact with extension stop at the end of swing



Prosthetic knees

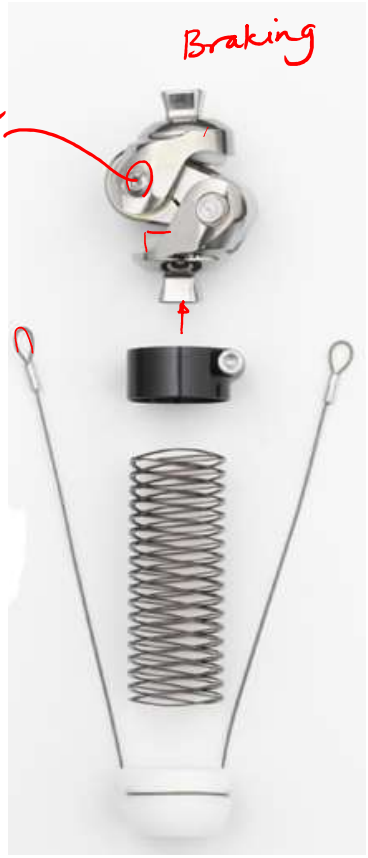
Geometric locking



Geoflex® Knee

Braking

Knee bolt



Stabilized® Knee

Single axis
- Swing control — Friction
  - Hydraulic
  - Pneumatic
  - MR fluid
  - Microprocessor control



Prosthetic knees



Fillauer Endurance 160
Hydraulic Knee





Otto Bock C-Leg

Prosthetic knees



OB 3R65



OB 3R78



OB 3R80